

## Solution

**Note:** The "soundness" from the notes refers to 1-soundness, but the one in the paper instead refers to  $N$ -soundness for each  $N$  in the first case the theorem is incorrect, so I assume that we're talking about the  $N$ -soundness for each  $N$ .

Let  $N_1 = (P_1, T_1, F_1, i_1, o_1)$  and  $N_2 = (P_2, T_2, F_2, i_2, o_2)$  be sound WF-nets. Assume  $P_1 \cap P_2 = \emptyset$  and  $T_1 \cap T_2 = \emptyset$  (disjoint sets of nodes). Let  $N' = (P', T', F', i', o')$  be the WF-net constructed by substituting place  $p \in P_1$  with the entirety of  $N_2$ .

Check that the net satisfies the definition of soundness:

- For each token put in pin, one and only one token eventually appears in pout,
- When the token appears in pout, all other places are empty, and
- For each transition (i.e, task), it is possible to move from the initial state to a state in which this transition is enabled.

The third condition is obvious from the fact that both nets are sound: for any marking, the tokens inside the subnet can be moved to the end since it's sound. And if they are at the "out" of the subnet, we can obtain a sequence of transitions that moves them to the "out".